

AP Statistics FRQ Practice Exam 04

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AP Statistics FRQ Practice Exam 4

APSTAT-FRQ4-001

Section: FRQ

Type: frq

Primary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Secondary Unit: None

FRQ Type: Formulate Questions and Collect Data

Prompt:

A language-learning app company wants to study weekly practice time among its 60,000 active adult users. The company takes a simple random sample of 800 users and records whether each user had voluntarily enabled a daily reminder before the study began. It then records each user's total practice time during the following week.

Reminder status	n	Mean weekly practice time	Standard deviation
Enabled	480	154 minutes	62 minutes
Not enabled	320	101 minutes	55 minutes

- (a) Identify the population and sample. Also identify the explanatory and response variables.
- (b) State a statistical question that the study can address. Calculate and interpret the observed difference in sample mean weekly practice time.
- (c) Identify the study as observational or experimental. Explain why the difference in means does not establish that enabling reminders causes an increase in practice time.
- (d) Describe one plausible confounding variable and explain how it could account for at least part of the observed association.
- (e) Explain what the simple random sample permits the company to generalize and what the study design does not permit the company to conclude.

Scoring Guide:

Total Points: 10

(a) 2 points

· Point 1: Identifies the population as all 60,000 active adult users and the sample as the 800 users selected by the SRS.

· Point 2: Identifies reminder status as the categorical explanatory variable and total weekly practice time in minutes as the quantitative response variable.

(b) 2 points

· Point 1: States an appropriate statistical question about whether weekly practice time differs or is associated with voluntarily enabled reminder status among active adult users.

· Point 2: Calculates $154 - 101 = 53$ minutes and interprets it: sampled users who had enabled reminders practiced an average of 53 minutes more during the week than sampled users who had not enabled reminders.

(c) 2 points

· Point 1: Identifies the study as observational because users chose their own reminder status and no treatment was randomly assigned.

· Point 2: Explains that the groups may differ in ways other than reminder status, so the observed association cannot isolate a causal effect of the reminders.

(d) 2 points

· Point 1: Identifies a plausible confounder, such as prior motivation, available free time, language-learning goals, or previous practice habits.

· Point 2: Explains how the selected confounder could be associated both with choosing to enable reminders and with spending more time practicing, offering an alternative explanation for the 53-minute difference.

(e) 2 points

· Point 1: Explains that the SRS supports generalizing the observed association to the company's 60,000 active adult users, assuming complete measurement and response.

· Point 2: States that the observational design does not support a cause-and-effect conclusion and that the result should not automatically be generalized to inactive users, children, or users of other apps.

Model Guidance:

The sample was randomly drawn from the company's active adult-user population, but reminder status was not assigned. The question is therefore about an association between a categorical explanatory variable and a quantitative response. The sample difference is 53 minutes, with the reminder-enabled group having the larger mean.

Random sampling and random assignment serve different purposes. The SRS supports extending the observed association to active adult users of this app. It does not make the two reminder groups comparable with respect to motivation or other preexisting characteristics. For example, highly motivated users may be more likely both to activate reminders and to practice longer, creating confounding. The data do not justify saying that reminders caused the difference.

Expected Points:

10

APSTAT-FRQ4-002

Section: FRQ

Type: frq

Primary Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Secondary Unit: None

FRQ Type: Analyze Data, One- or Two-Variable Data and Probability

Prompt:

Wildlife biologists evaluate an acoustic classifier that flags audio segments as containing a possible canyon bat call. Human experts independently determine whether a canyon bat call is actually present. Results for 9,500 archived audio segments are shown.

	Classifier flagged	Classifier did not flag	Total
Canyon bat call present	168	42	210
Canyon bat call absent	392	8,898	9,290
Total	560	8,940	9,500

Suppose one of the 9,500 segments is selected at random.

- (a) Calculate the probability that the classifier flagged the segment and the probability that a canyon bat call is actually present.
- (b) Calculate the probability that the segment was flagged or actually contains a canyon bat call. Interpret the result.
- (c) Given that the classifier flagged the segment, calculate the probability that a canyon bat call is actually present. Interpret the result.
- (d) Given that a canyon bat call is actually present, calculate the probability that the classifier flagged the segment. Explain why this result differs from the result in part (c).
- (e) Determine whether the events “classifier flagged the segment” and “canyon bat call is present” are independent. Justify your answer with probabilities from the table, and evaluate the claim, “A flagged segment has an 80% chance of containing a canyon bat call.”

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Calculates $P(\text{flagged}) = 560/9,500 \approx 0.0589$.
- Point 2: Calculates $P(\text{call present}) = 210/9,500 \approx 0.0221$.

(b) 2 points

- Point 1: Calculates $P(\text{flagged or call present}) = (560 + 210 - 168)/9,500 = 602/9,500 \approx 0.0634$.
- Point 2: Interprets the result: about 6.34% of the archived segments were flagged, actually contained a canyon bat call, or both.

(c) 2 points

- Point 1: Calculates $P(\text{call present} \mid \text{flagged}) = 168/560 = 0.300$.
- Point 2: Interprets the result: among flagged segments, 30.0% actually contain a canyon bat call.

(d) 2 points

- Point 1: Calculates $P(\text{flagged} \mid \text{call present}) = 168/210 = 0.800$.
- Point 2: Explains that the conditioning groups differ: part (c) uses the 560 flagged segments as the denominator, whereas part (d) uses the 210 segments that actually contain a call.

(e) 2 points

- Point 1: Concludes that the events are not independent because $P(\text{call present} \mid \text{flagged}) = 0.300$ differs from $P(\text{call present}) \approx 0.0221$; an equivalent valid probability comparison earns the point.
- Point 2: Rejects the claim because 80% is the probability of a flag given a true call, not the probability of a true call given a flag; the latter is only 30%.

Model Guidance:

The table distinguishes two conditional probabilities that are often confused. The classifier flags 560 of 9,500 segments, while only 210 of 9,500 actually contain the call. For the union in part (b), subtract the 168 segments counted in both events once. The conditional probability given a flag is $168/560 = 0.300$. Reversing the condition gives $168/210 = 0.800$, which measures how often the system detects a true call.

The flag and true-call events are associated because a flagged segment is much more likely to contain a call than a randomly selected segment: 0.300 versus about 0.0221. Still, the rarity of actual calls means most flags are false positives. Therefore, an 80% detection rate does not imply that 80% of flags are genuine calls.

Expected Points:

10

APSTAT-FRQ4-003

Section: FRQ

Type: frq

Primary Unit: Unit 3, Inference for Categorical Data: Proportions

Secondary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

FRQ Type: Analyze Data, Statistical Inference (Confidence Interval or Significance Test)

Prompt:

A statewide association maintains a list of 2,800 independently owned bookstores. It takes a simple random sample of 240 stores and finds that 146 offer same-day pickup for online orders.

Construct and interpret a 95% confidence interval for the true proportion of all independently owned bookstores on the association's statewide list that offer same-day pickup. Use $z^* = 1.96$.

- Define the population parameter and identify the appropriate confidence-interval procedure.
- Check the conditions for the confidence-interval procedure.
- Calculate the sample proportion, standard error, and margin of error, and construct the confidence interval.
- Interpret the 95% confidence interval in context.

(e) Use the interval to evaluate the claim that a majority of the bookstores on the statewide list offer same-day pickup.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Defines p as the true proportion of all 2,800 independently owned bookstores on the statewide list that offer same-day pickup.
- Point 2: Selects a one-sample z confidence interval for a population proportion.

(b) 3 points

- Point 1: Cites the simple random sample.
- Point 2: Verifies the 10% condition because $240 \leq 0.10(2,800)$.
- Point 3: Verifies the large-counts condition using 146 successes and $240 - 146 = 94$ failures, both at least 10.

(c) 3 points

- Point 1: Calculates $\hat{p} = 146/240 \approx 0.6083$ and $SE = \sqrt{[0.6083(1 - 0.6083)/240]} \approx 0.03151$.
- Point 2: Calculates the margin of error as $1.96(0.03151) \approx 0.06176$.
- Point 3: Constructs 0.6083 ± 0.0618 , giving the interval $(0.5466, 0.6701)$, allowing for minor rounding differences.

(d) 1 point

- Point 1: States that the association is 95% confident that between about 54.7% and 67.0% of all independently owned bookstores on its statewide list offer same-day pickup.

(e) 1 point

- Point 1: Supports the majority claim because every plausible value in the 95% confidence interval is greater than 0.50.

Model Guidance:

The parameter is the statewide-list proportion, not the sample proportion. A one-proportion z interval is appropriate because the stores were randomly sampled, 240 is less than 10% of 2,800, and the sample includes well over 10 stores in each outcome category.

The estimate is $146/240 \approx 0.6083$. Its estimated standard error is approximately 0.03151, so the 95% margin of error is about 0.06176. The resulting interval is approximately $(0.5466, 0.6701)$. This interval estimates the proportion only for bookstores on the association's list. Since the entire interval lies above 0.50, it provides evidence that a majority of the listed bookstores offer same-day pickup.

Expected Points:

10

APSTAT-FRQ4-004

Section: FRQ

Type: frq

Primary Unit: Unit 5, Regression Analysis

Secondary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

FRQ Type: Interpret Results, Integrated Analysis and Justification

Prompt:

A school district examines the association between x , average weekly minutes of school-provided tutoring per student, and y , the percentage of students passing a common end-of-year mathematics assessment. Data for five middle schools and five high schools are shown.

School	Level	Tutoring minutes, x	Pass rate, y
M1	Middle	20	84%
M2	Middle	25	85%
M3	Middle	30	86%
M4	Middle	35	88%
M5	Middle	40	89%
H1	High	60	70%
H2	High	65	71%
H3	High	70	73%
H4	High	75	74%
H5	High	80	76%

Regression output:

All 10 schools: $\hat{y} = 93.156 - 0.271x$, $r = -0.811$

Middle schools: $\hat{y} = 78.600 + 0.260x$, $r = 0.991$

High schools: $\hat{y} = 51.800 + 0.300x$, $r = 0.993$

- (a) On paper, sketch a scatterplot of pass rate against tutoring minutes, using different symbols for middle and high schools. Label both axes. Describe the association for all schools together and within each school level.
- (b) Interpret the slope of the regression line for all 10 schools. Use that line to predict the pass rate for a school averaging 50 tutoring minutes per student per week, and explain one reason to treat this prediction cautiously.
- (c) Explain how the association can be negative for all schools combined but positive within both school levels. Identify the role of school level in this pattern.
- (d) A district administrator claims, "These data show that providing more tutoring lowers mathematics pass rates." Evaluate the claim using the study design and the regression evidence.
- (e) For a high school averaging 70 tutoring minutes, calculate the predicted pass rate from both the all-school line and the high-school line. Decide which prediction is more appropriate and justify your choice.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Sketches all 10 points with tutoring minutes on the horizontal axis and pass rate on the vertical axis, uses distinguishable symbols for the two school levels, and includes appropriate labels and scales.
- Point 2: Describes a moderately strong negative linear association when all schools are combined but very strong positive linear associations within both the middle-school and high-school groups.

(b) 2 points

- Point 1: Interprets the combined slope: across these 10 schools, each additional average tutoring minute per student is associated with a predicted decrease of about 0.271 percentage point in pass rate, on average.
- Point 2: Calculates $\hat{y} = 93.156 - 0.271(50) = 79.606\%$, and notes that 50 minutes lies in a gap between the observed middle- and high-school x-values, where the combined line ignores school level.

(c) 2 points

- Point 1: Explains that high schools have both more tutoring minutes and lower baseline pass rates than middle schools, so combining the separated groups produces a negative overall trend even though each within-level trend is positive.
- Point 2: Identifies school level as a lurking or confounding variable associated with both average tutoring minutes and pass rate; aggregation masks the within-level relationships.

(d) 2 points

- Point 1: Rejects the causal claim because the data are observational and schools were not randomly assigned tutoring amounts; factors such as course difficulty, student age, prior achievement, or allocation of tutoring to higher-need schools could explain the association.
- Point 2: Uses the positive within-level slopes and correlations as evidence against the administrator's interpretation of the negative combined slope; the aggregated association does not show that additional tutoring lowers pass rates.

(e) 2 points

- Point 1: Calculates the all-school prediction as $93.156 - 0.271(70) = 74.186\%$ and the high-school prediction as $51.800 + 0.300(70) = 72.800\%$.
- Point 2: Selects the high-school prediction as more appropriate because it is based on the relevant subgroup, 70 minutes lies within the observed high-school range, and the subgroup relationship avoids the distortion caused by combining school levels.

Model Guidance:

The scatterplot should reveal two distinct clusters. Within each cluster, pass rate rises as tutoring time rises. Because the high-school cluster is farther right and lower than the middle-school cluster, the combined cloud slopes downward. This reversal is an aggregation effect caused by school level, which is related to both variables.

The combined line predicts about 79.6% at 50 minutes, but no observed school is near 50 minutes and the two levels occupy different ranges. The negative combined slope is an association, not an effect of assigning more tutoring. The study is observational, and the positive subgroup slopes directly undermine a causal reading of the combined slope. For a 70-minute high school, the subgroup prediction of 72.8% is preferable to the combined prediction of about 74.2% because it uses the relevant within-level pattern. Even the subgroup line remains descriptive rather than causal.

Expected Points:

10

SERIES DO-NOT-REPEAT BANK UPDATE, APSTAT-FRQ4

- APSTAT-FRQ4-001: Unit 1 primary; no secondary unit; observational comparison from an SRS of 800 of 60,000 active adult language-app users who voluntarily enabled or did not enable daily reminders; summary table $n = 480$ versus 320, mean weekly practice 154 versus 101 minutes, SD 62 versus 55; formulate an association question, calculate and interpret a 53-minute mean difference, identify motivation or related user traits as confounders, and separate random-sampling generalizability from the lack of causal inference; targets treating self-selected exposure groups as randomized treatments.
- APSTAT-FRQ4-002: Unit 2 primary; no secondary unit; two-way table for 9,500 archived acoustic segments cross-classified by expert-confirmed canyon bat call and classifier flag, with cells 168, 42, 392, and 8,898; calculate marginal and union probabilities, $P(\text{call} \mid \text{flagged}) = 0.300$, $P(\text{flagged} \mid \text{call}) = 0.800$, assess independence, and reject the reversal of sensitivity into positive predictive value; targets confusing conditional direction and ignoring a low base rate.
- APSTAT-FRQ4-003: Unit 3 primary; Unit 1 secondary; one-proportion z confidence interval for same-day pickup among 2,800 independently owned bookstores on a statewide association list, using an SRS of 240 with 146 successes; define p , check SRS/10%/large-count conditions, calculate $\hat{p} \approx 0.6083$, $SE \approx 0.03151$, 95% margin of error ≈ 0.06176 , interval (0.5466, 0.6701), interpret scope, and support a majority claim; targets confusing p with $\hat{p}$ and omitting population scope from interval interpretation.
- APSTAT-FRQ4-004: Unit 5 primary; Unit 1 secondary; raw table for five middle and five high schools relating average tutoring minutes to mathematics pass rate; combined regression $\hat{y} = 93.156 - 0.271x$ with $r = -0.811$ versus positive within-level lines and correlations; create a symbol-coded scatterplot, interpret and cautiously use the combined line in the between-group gap, explain association reversal from aggregation and school-level confounding, reject a causal “tutoring lowers pass rates” claim, and compare all-school versus high-school predictions at $x = 70$; targets ignoring subgroup structure, treating aggregate correlation as causal, and choosing a combined model when a relevant subgroup model exists.